

DAY 2 MINI PROJECT

ENTERPRISE PUBLIC VS PRIVATE IP ADDRESS ANALYSIS

Project Objective

The objective of this project is to analyze public and private IP addresses, understand NAT communication, and explore how organizations use IP-based security controls in enterprise IAM environments.

This project also demonstrates practical networking analysis using real-world troubleshooting and IAM security concepts.

Introduction

Every device connected to a network requires an IP address for communication.

IP addresses are categorized into:

1. Public IP Address
2. Private IP Address

Understanding these concepts is essential in:

- Networking
- Cybersecurity
- Cloud Security
- Identity & Access Management (IAM)

Private IP Address Analysis

My Private IP Address

My private IP

10.221.64.26

Explanation

A private IP address is used inside local/internal networks.

Private IP addresses:

- Are not directly visible on the internet
- Are assigned internally by routers
- Help organizations securely manage internal communication

Private IP Ranges

10.0.0.0 – 10.255.255.255

172.16.0.0 – 172.31.255.255

192.168.0.0 – 192.168.255.255



Public IP Address Analysis

My Public IP Address

My public IP)

Example:

106.192.36.233

ISP Information

My ISP name is:

Bharti Airtel Ltd.

Explanation

A public IP address is internet-visible and assigned by the Internet Service Provider (ISP).

Public IP addresses allow communication between local networks and the internet.



NAT (Network Address Translation)

NAT converts:

Private IP → Public IP

Example:

10.221.64.26

↓

106.192.36.233



Why NAT Is Important

NAT helps:

- Save IPv4 addresses
- Improve security
- Hide internal devices
- Allow internet communication



Network Communication Flow

Laptop

↓

Private IP
↓
Router/Gateway
↓
NAT Translation
↓
Public IP
↓
ISP
↓
Internet

❏ DNS Analysis

DNS (Domain Name System) converts domain names into IP addresses.

Example:

google.com → IP address

This helps users access websites using names instead of numerical IP addresses.

⌘ Commands Used

ipconfig

ping google.com

nslookup google.com

🔑 Ping Analysis

The ping command was used to:

- Test internet connectivity
- Measure response time
- Verify communication with external servers

🔑 DNS Lookup Analysis

The nslookup command was used to:

- Analyze DNS resolution
- Convert domain names into IP addresses

🔒 IAM Security Mapping

Organizations use IP addresses in IAM systems for:

- Conditional Access

- Geo-location tracking
- MFA triggering
- Suspicious login detection
- VPN restrictions
- Risk-based authentication

Enterprise Scenario Analysis

Scenario

An employee usually logs in from India.

Suddenly, a login attempt occurs from another country using an unknown public IP address.

IAM Security Response

IAM systems may:

- Trigger Multi-Factor Authentication (MFA)
- Block the login
- Alert security teams
- Mark login as risky
- Apply conditional access policies

Zero Trust Security Concept

Modern organizations follow:

“Never Trust, Always Verify”

Even if login comes from a valid IP:

- Identity verification is still required
- Device trust is validated
- Risk analysis is performed

Firewall & Security Controls

Organizations use firewalls to:

- Allow trusted IP addresses
- Block suspicious IP addresses
- Restrict traffic from risky countries
- Monitor network communication

Public IP

Visible on internet

Assigned by ISP

Globally unique

Used for internet communication Used for internal communication

Private IP

Used inside local network

Assigned by router

Internal use

Used for internal communication

1. Private IP (ipconfig)

[illegible]

2. Public IP lookup

The screenshot shows the homepage of WhatIsMyIPAddress.com. The navigation bar includes links for ABOUT, PRESS, BOOK, PODCAST, and CONTACT. Below the navigation bar, there are tabs for MY IP, IP LOOKUP (selected), HIDE MY IP, VPNs, TOOLS, and LEARN. The main content area displays the user's public IP address: IPv6: 2401:4900:97c2:aa1e:81d:73e6:3dec:8be0 and IPv4: 106.192.36.233. It also shows the user's location as Visakhapatnam, India, with a map. A red button labeled 'HIDE MY IP ADDRESS NOW' is visible. The page also features a Canva advertisement and a warning about location exposure.

3. Ping command output

```
Microsoft Windows [Version 10.0.26200.8524]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Akshaya>ping google.com

Pinging google.com [2404:6800:4007:833::200e] with 32 bytes of data:
Reply from 2404:6800:4007:833::200e: time=58ms
Reply from 2404:6800:4007:833::200e: time=115ms
Reply from 2404:6800:4007:833::200e: time=92ms
Reply from 2404:6800:4007:833::200e: time=51ms

Ping statistics for 2404:6800:4007:833::200e:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 51ms, Maximum = 115ms, Average = 79ms

C:\Users\Akshaya>
```

4. DNS lookup output

```
Microsoft Windows [Version 10.0.26200.8524]
(c) Microsoft Corporation. All rights reserved.

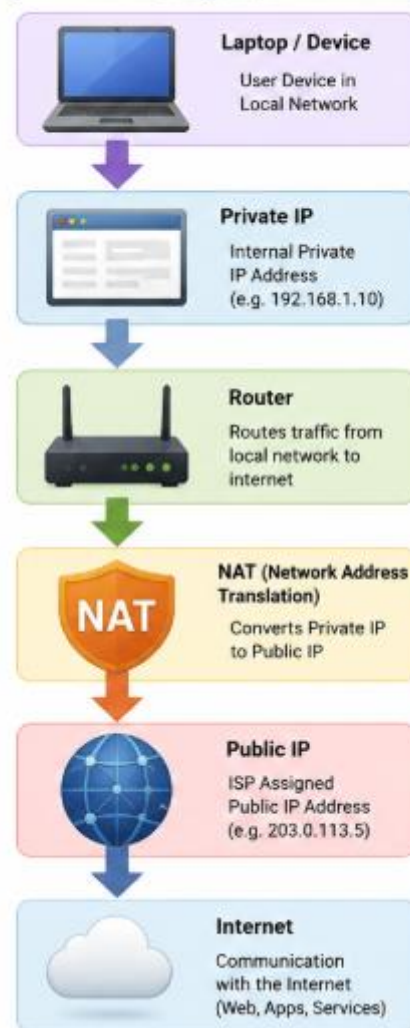
C:\Users\Akshaya>nslookup google.com
Server: UnKnown
Address: 10.221.64.104

Non-authoritative answer:
Name: google.com
Addresses: 2404:6800:4007:801::200e
          142.250.205.46

C:\Users\Akshaya>
```

5. NAT flow/network diagram

NAT FLOW DIAGRAM



Key Learning Outcome

Through this project, I learned:

- Difference between public and private IP addresses
- NAT communication flow
- DNS resolution process
- Enterprise networking concepts
- IAM security mapping using IP addresses
- Conditional access and geo-location security concepts

This project strengthened my foundational skills in:

- Networking
- Cybersecurity
- IAM
- Cloud Security
- Security Operations